Herded or Anchored? Testing Conformity in Machine Learning on a Decade of ICLR Submissions

by

Minghan Li

A report submitted for
Science Research Project (SCIE30001)

Supervisor: A/Prof Jianzhong Qi

in the
School of Computing and Information Systems
Faculty of Engineering and Information Technology
THE UNIVERSITY OF MELBOURNE

November 2026

THE UNIVERSITY OF MELBOURNE

Abstract

School of Computing and Information Systems
Faculty of Engineering and Information Technology

Science Research Project (SCIE30001)

by Minghan Li

Machine learning is often said to herd: researchers chase what is popular, and peer review rewards them for it. But a field can converge without copying. Researchers may instead stay close to a record: their own earlier work, or a topic they already know. We call these two routes to conformity *imitation* and *continuity*, and test the herding account where they make different predictions. The data are 43,480 ICLR submissions from 2017 to 2026, including rejected and withdrawn papers; hypotheses were registered before any outcome was examined and re-tested on 11,099 held-out papers. The field did converge on foundation models, but both routes predict this, and the number of topics that appear to be emerging depends on hindsight: between 8 and 29, depending on which years the topic model is fitted to. Where the two routes make different predictions, we find no evidence of herding. Researchers new to a topic propose methods more often than incumbents do, the reverse of our registered prediction. At review, acceptance shows no detectable difference for papers in crowded topics, whereas papers far from their authors' earlier work are accepted less often (a post hoc finding, confirmed on held-out papers). The conformity visible at ICLR is anchored to researchers' own records rather than driven by the crowd.

Declaration of Authorship

I, Minghan Li, declare that this report, titled 'Herded or Anchored? Testing Conformity in Machine Learning on a Decade of ICLR Submissions', and the work presented in it are my own. I confirm that:

- the report comprises only my original work towards Science Research Project (SCIE30001) except where indicated in the preface;
- due acknowledgement has been made in the text to all other material used; and
- the report is within the word limit set for it, exclusive of tables, figures and the bibliography.

Signed: _____

Date: _____

Preface

This report is my own work for Science Research Project (SCIE30001), carried out under the supervision of A/Prof Jianzhong Qi. It has not been submitted for any other qualification, and it has not been published or submitted for publication.

Acknowledgements

I thank my supervisor, A/Prof Jianzhong Qi, for guidance and feedback throughout this project.

Contents

Abstract i

Declaration of Authorship ii

Preface iii

Acknowledgements iv

List of Figures vii

List of Tables viii

Abbreviations ix

Symbols x

1 Introduction 1

2 Two Routes to Conformity 4
2.1 Imitation . . . 4
2.2 Continuity . . . 4
2.3 Rival explanations and the timing of predictions . . . 5

3 Data and Design 6
3.1 Corpus . . . 6
3.2 Instruments . . . 6
3.3 Entry profiles . . . 7
3.4 Two distances . . . 7
3.5 Prospective measurement . . . 7
3.6 Protocol . . . 8

4 The Field: What Is Fashion and What Lasts 9
4.1 Waves . . . 9
4.2 Emergence depends on hindsight . . . 9
4.3 What lasts: the shift to foundation models . . . 10

5 Entry: Who Writes What **11**
5.1 Newcomers propose methods 11
5.2 Topic newcomers and first-time ICLR teams 12
5.3 Farther from earlier work 13
5.4 Later uptake of foundation models 14
5.5 Open questions 14

6 The Gate: Crowd or Record? **15**
6.1 Is following the crowd associated with acceptance? 15
6.2 What is acceptance associated with? 16
6.3 Interpretation 19

Bibliography **20**

List of Figures

1.1 Keyword trends at ICLR, 2018–2026 1

4.1 Emerging topics by fitting window, and the shift to foundation models . . 10

5.1 Entry contrasts within topic and year 13

6.1 Gate tests by reference point 18

List of Tables

2.1 What each route predicts for the field, at entry and at the gate 5

5.1 Entry contrasts between newcomer and incumbent papers 12

6.1 Acceptance contrasts at the gate 17

Abbreviations

CI	Confidence interval (95% unless stated)
FM	Foundation model
ICLR	International Conference on Learning Representations
LLM	Large language model
MDE80	Minimum detectable effect at 80% power
ML	Machine learning
NLP	Natural language processing
pp	Percentage points

Symbols

$F_{\max}$	closeness to the team's earlier ICLR work	cosine similarity
k	number of topics in a topic map	–
κ	Cohen's κ, agreement between two labelling passes	–

Chapter 1

Introduction

Machine learning (ML) is often said to herd. Between 2017 and 2026, yearly submissions to the International Conference on Learning Representations (ICLR) in our discovery set (Chapter 3) grew 40-fold, from 384 to 15,411. Over the same years, the share of submissions building on a foundation model (FM) rose from 0.3% to 54.4%, while the share proposing a new architecture fell from 24.5% to 7.5%. Individual keywords rose and fell (Figure 1.1): generative adversarial networks declined from 8.1% of submissions in 2018, and large language models (LLMs) reached 16.7% in 2026. The rise of LLMs also involves natural language processing (NLP) researchers, who increasingly publish at ML venues [1]. Whether science has fashions is an old question [2]. Critics of ML

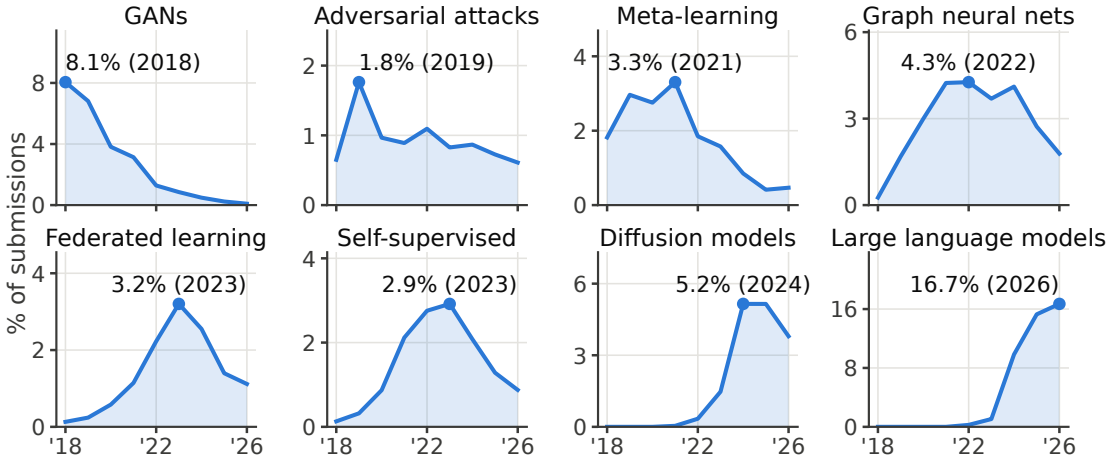


FIGURE 1.1: Keyword trends at ICLR: share of submissions in the discovery set listing each author keyword, 2018–2026; dots mark the peak. The eight keywords were chosen for display after their curves were known. Across the 68 most frequent keywords, 17 peaked in 2019–2024 with shares of at most half the peak both before it and in 2026; Chapter 4 calls these waves.

scholarship point to incentives that favour short-term success [3] and to convergence on whatever suits the field's hardware, benchmarks and foundation models [4–6], while cascade and fashion theories explain waves by people taking their cue from others [7–9].

Is the switch to FMs not itself herding? Not necessarily. Lock-in, the efficient choice of a tool that works and a bandwagon all predict convergence on a shared tool, so the rising FM share (Figure 4.1b) cannot tell them apart. The herding question is about behaviour: what researchers write when they enter a topic, and what peer review accepts. Here two routes to conformity diverge. *Imitation* takes its reference point from what others are doing now: what is popular, rising or recently accepted. *Continuity* takes it from a record: the team's own earlier work, or a topic that a team member already knows. Imitation makes teams alike, whereas continuity keeps each team near its own past. We call work that conforms by continuity *anchored*.[1]

Read as herding, the waves make three claims: they are fashions; newcomers to a topic carry them by following others; and review favours following. We test each claim at the level where it applies: in the field as a whole, at entry to a topic, and at the acceptance decision, which we call the gate. The data are 43,480 ICLR submissions from 2017 to 2026, including rejected and withdrawn papers. Hypotheses were registered before any paper in the full corpus was labelled, and outcomes were examined only once, in a single pre-specified analysis run. The main tests were then run again on 11,099 held-out papers. This is a pre-registered out-of-sample replication, not a certified untouched holdout. Each finding carries one of three labels. A REPLICATED finding comes from a registered test whose discovery result, in either direction, was confirmed on held-out papers by a test registered before the held-out data were read. A HELD-OUT SUPPORTED finding was found post hoc and then supported by a test declared before the held-out data were read. An EXPLORATORY finding rests on the discovery papers alone or is descriptive. Tests that reach no verdict are called inconclusive or uninformative, as defined in Chapter 3.

In the field as a whole, keyword waves are real, but the number of topics that appear to be emerging depends on hindsight, and the one durable shift, the move to foundation models, fits both routes. At entry, researchers new to a topic propose methods more

[1] 'Anchored' describes a position relative to a record and implies no particular mechanism. It does not refer to the anchoring-and-adjustment heuristic [10], the sense in which writing on herding in science uses the word [11].

often than incumbents do, and in 2024–2026, the period we call the FM era, measurement papers (those contributing a benchmark, dataset or evaluation) are concentrated among incumbents and larger teams (EXPLORATORY). Both patterns run against herding, which implies that newcomers write follow-on work (benchmarking or evaluating others' methods), and against our own registered prediction. At the gate, we find no evidence of a premium for crowded topics. Titles that open with a coined name carry a small premium, but the held-out test lacked the power to confirm it, and most of what held up on held-out papers is tied to the team's own earlier work.

Prior work relates ICLR acceptance to prestige, process and research area [12–16], and describes which papers start topics, who enters LLM research and how far researchers move from their own past work [17–19]. To our knowledge, no study has tested the herding account both at entry and at the gate within one submission population, using registered hypotheses that are re-tested on held-out papers, an approach Ibrahim et al. [13] applied to a prestige gap. This report makes three contributions.

- First, at entry, we report a registered test of herding where imitation predicts it most sharply. Researchers entering a topic propose methods more often than incumbents, by 3.7 percentage points (pp), a REPLICATED result. The surplus holds for researchers with ICLR work in other topics and appears in every period, including before 2023.
- Second, at the gate, using decided submissions under double-blind review, rejected ones included, we find that acceptance is associated with closeness to the team's own earlier work: papers far from it are accepted 5.4 pp less often in discovery and 6.1 pp less often on held-out papers (HELD-OUT SUPPORTED). We find no evidence that following the crowd is associated with acceptance. Hill et al. [20] relate moving away from one's own research to publication success; we add the accept/reject decision.
- Third, on method, we show that the same papers yield between 8 and 29 emerging topics, depending on the years the topic model is fitted to (EXPLORATORY). Emergence is hard to judge before the fact [21, 22], and a prospective signal has been tested [23]; refitting the topic model on early years is a cheap check on claims about trends.

Chapter 2 sets out the two routes and what each predicts. Chapter 3 describes the data, measures and testing protocol. Chapters 4, 5 and 6 report results for the field, for entry to topics and for the acceptance decision.

Chapter 2

Two Routes to Conformity

2.1 Imitation

When outcomes are uncertain, organisations mimic one another [24], individuals follow their predecessors and discard their own information [7, 8], and herding protects a reputation when evaluators judge people against their peers [25]. Bandwagons can spread even when adopting brings no gain [26]. In ML, benchmark use is increasingly concentrated on a small set of datasets [27]. Entrants to a topic face the most uncertainty, so imitation predicts that they flock to hot topics, adopt rising tools early and write follow-on work that benchmarks or evaluates what others have proposed. If review favours following, imitation also predicts that, at the gate, papers in crowded topics and papers with branded titles (a short coined name and a colon) or keywords near their peak are accepted more often.

2.2 Continuity

Search stays near existing competence [28, 29], and taking up new knowledge depends on related prior knowledge [30]. At entry, continuity predicts that incumbents consolidate a topic's existing body of work and that new tools spread through teams that already have earlier ICLR work. Entrants, on this view, bring work from elsewhere rather than copy what a topic already does, as accounts of outsiders in science expect [31, 32]. At the gate, acceptance should track how close a paper stays to its team's record. Weisberg and

Route (reference point)	Field	Entry	Gate
Imitation (others, now)	Convergence on shared tools	Entrants flock to hot topics, adopt rising tools early and write follow-on work that benchmarks or evaluates others' methods	Acceptance responds to crowding (the registered test predicted a penalty); a premium for branded titles and for keywords near their peak
Continuity (own record)	Convergence on shared tools	Entrants bring work from other topics; incumbents consolidate; new tools spread through teams with earlier ICLR work	Acceptance tracks closeness to the team's own record

TABLE 2.1: What each route predicts at the three levels. Lock-in and efficient choice also predict convergence on shared tools, so only entry and the gate can tell the routes apart.

Muldoon [33] separate 'followers' of others' approaches from agents who ignore them; we borrow the distinction but not their welfare conclusions, which Alexander et al. [34] dispute.

2.3 Rival explanations and the timing of predictions

Visible status, cumulative advantage and expertise [13, 35, 36] predict as well that acceptance is associated with a team's record. Network diffusion, in which less connected actors adopt later [37], makes the same prediction as continuity for new tools, namely that teams new to ICLR take them up later, so that contrast cannot separate the routes.

The hypotheses we registered before reading any outcome were imitation's entry predictions and the crowding test at the gate (Table 2.1). The registered crowding test predicted a penalty for crowded topics, whereas imitation, if review rewards it, would predict a premium; a result of either sign would therefore show review responding to the crowd. Continuity's predictions were formulated after the tests had been run and are used here only to organise the results.

Chapter 3

Data and Design

3.1 Corpus

The corpus was built from official OpenReview submission data, which we downloaded ourselves on documented dates. The discovery set (43,480 submissions with a usable abstract) was used to develop and run the registered tests. The held-out set (11,099 papers) consists of submissions that the data source had reserved for confirmation; they were released to us only after the discovery tests had been registered and run. Acceptance tests use the 32,190 decided discovery papers submitted in 2018–2026 (2017 serves only as earlier history), of which 40.1% were accepted, and the 8,185 decided held-out papers from the same years; withdrawn papers are left out. The field and entry analyses use no acceptance decisions.

3.2 Instruments

A *topic map* is one spherical k-means clustering ($k = 100$) of SPECTER2 embeddings of title and abstract [38, 39], fitted on a given set of years; a *topic* is one cluster. Unless another map is named, topics come from the map fitted on all years. An LLM with a fixed codebook assigns each paper one primary contribution type. A *method paper* proposes a method, including a new architecture, and a *measurement paper* contributes a dataset, benchmark or evaluation. An *FM paper* is one that uses or adapts a large pre-trained model. A blind second labelling pass agreed with the first at Cohen's $\kappa = 0.93$,

and we hand-checked a small random sample. This measures consistency between two LLM passes; it does not establish agreement with human coders, which for zero-shot LLM labels is often only fair [40].

3.3 Entry profiles

Entry status is defined relative to a topic and counts ICLR submissions only. In a *newcomer paper*, no author has an earlier ICLR submission in the paper's topic, whatever its outcome (a topic-relative definition, as in Biliotti et al. [41]); otherwise it is an *incumbent paper*, and an author with such a submission is a *topic insider*. A newcomer paper comes from a *first-time team* if no author has any earlier ICLR submission, and from a *returning topic newcomer* if some author has one in another topic.

3.4 Two distances

Both distances are measured in one centred SPECTER2 embedding space, but from different reference points. *Closeness to the team's own record*, $F_{\max}$, is the highest cosine similarity between the paper and the mean of an author's earlier ICLR submissions since 2021, taken over the paper's *experienced authors* (those with such submissions), following Bao and Li [19]. *Distance from earlier ICLR work* is one minus the mean cosine similarity to the ten nearest submissions from earlier years, expressed as a percentile within topic and year.

3.5 Prospective measurement

Predictors use only information from the past. *Prior FM adoption* is the FM share of the topic's papers in the previous year, *crowding* is the log of one plus the number of the topic's earlier papers, and a *hot topic* is one in the top fifth of prior growth. Links to code repositories, usually added after acceptance, appear in 17.6% of accepted but only 0.5% of rejected 2023–2026 abstracts and so leak the outcome [42]. Removing them leaves the discovery estimates of the distance-based acceptance tests unchanged; the check is not possible for held-out papers.

3.6 Protocol

Hypotheses, models and a support rule were registered before the full corpus had been labelled. The registration and each later declaration are dated documents kept with the analysis code. Each registered test was re-run under pre-specified variants (other topic maps and numbers of topics, a year-balanced map, subsets of years and of papers); tests were grouped into families by research question for the Holm correction [43]. A registered hypothesis is supported if its estimate has the predicted sign with a Holm-adjusted $p < 0.05$ in its family, and keeps that sign in at least 75% of the counted variants and in both hindsight variants (the year-balanced map, and the 2017–2020 map applied to 2021–2026 papers). Outcomes were examined only by pre-committed analysis code, in a single logged run [44, 45].

Post hoc analyses were declared before they were run, and their one-sided held-out tests were declared before any held-out estimate for that question was computed. A declared held-out test is supported if its one-sided p is below 0.05; Holm-adjusted values across the six secondary held-out tests are reported alongside but do not set the label. Because held-out tests are one-sided, a supported held-out test confirms the sign of an association; its size should not be compared with the discovery estimate. A test that is not supported is *uninformative* when its estimate lies below its minimum detectable effect at 80% power (MDE80), and *inconclusive* otherwise or when no MDE80 is available. An uninformative test cannot distinguish a small association from none. Unless stated otherwise, models are linear probability models with topic and year fixed effects and errors clustered by topic.

Two limitations should be noted. First, under a simulated null whose structure cuts across topics, the main entry tests rejected 12.6% and 10.0% of the time at a nominal 5% level, and a wild cluster bootstrap [46] did not correct this. Second, keyword popularity and title style had been explored against acceptance on the discovery papers before registration. The discovery estimate for titles is therefore not a test, and the keyword-timing tests probe a dimension that had already been seen.

Chapter 4

The Field: What Is Fashion and What Lasts

This chapter asks which changes in the field are passing fashions and which persist.

4.1 Waves

Author keywords need no topic map, and at their level waves are real: 17 of the 68 most frequent keywords are *waves*, which rose and fell, peaking in 2019–2024 with shares of at most half the peak both before it and in 2026 (Figure 1.1; EXPLORATORY).

4.2 Emergence depends on hindsight

Histories of a field usually fit one topic model to the whole archive [47–49], as do dynamic topic models and BERTopic [50, 51]. We call a topic *emerging* if its 2024–2026 share is at least three times its 2017–2020 share, and *declining* if its 2024–2026 share is at most a third of its 2017–2020 share. With the same papers and pipeline, a map fitted on 2017–2020 finds 8 emerging topics and one fitted on all years finds 29 (Figure 4.1a; EXPLORATORY). Two causes combine, and we cannot separate them. The all-years map leaks the future into the past, like temporal leakage in prediction [52] and hindsight bias in judgement [53]: with 69% of its papers in 2024–2026, it is in effect a present-day

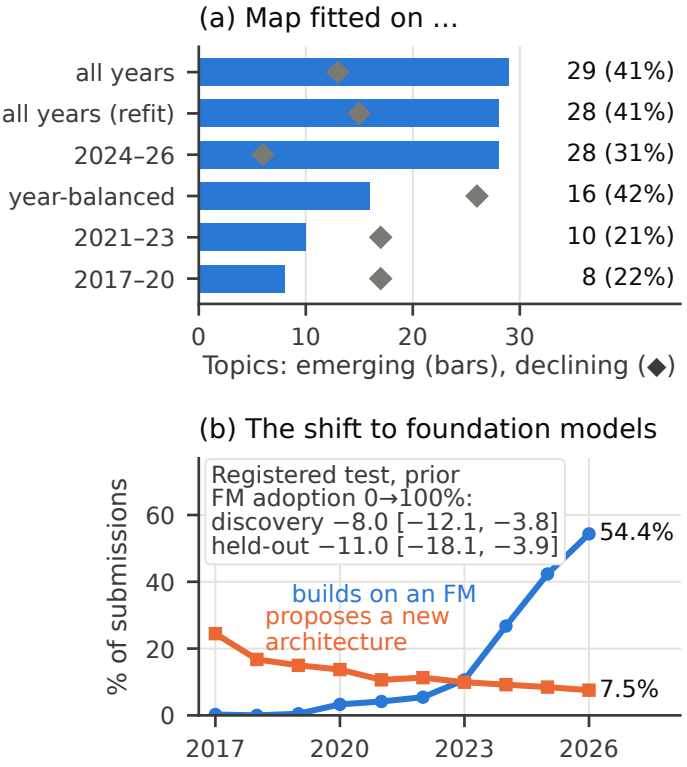


FIGURE 4.1: (a) Emerging (bars) and declining (grey diamonds) topics on topic maps fitted on different years, for the same 43,480 papers. Labels at right show the number of emerging topics and, in brackets, the share of 2024–2026 papers in them (22.3% on the 2017–2020 map, 40.9% on the all-years map); 'all years (refit)' uses another random seed. Topic size 'predicts' two-year growth only on the all-years map (out-of-sample $R^2 = 0.28$ against -0.03; no paired test). (b) Share of submissions building on an FM (blue) and proposing a new architecture (orange), using contribution types assigned by the LLM, with the registered test inset (REPLICATED).

map. The early map cannot represent topics born after 2020. The two counts therefore bracket the number of emerging topics.

4.3 What lasts: the shift to foundation models

The FM share rose every year from 2018 (Figure 4.1b). Prior FM adoption is associated with fewer new-architecture papers: from 0 to 100% adoption the difference is -7.98 pp $[-12.14, -3.82]$ in discovery and -11.0 $[-18.1, -3.9]$ held-out (REPLICATED), though the association may run the other way. The test for a parallel shift towards empirical or evaluation papers (a separate registered measure) is inconclusive (-0.63 $[-3.64, +2.38]$). The pattern matches accounts of NLP's paradigm shifts [54–56], but lock-in [4], efficient choice [57] and a bandwagon [58] predict it too, so only entry and the gate can separate the routes.

Chapter 5

Entry: Who Writes What

This chapter asks what researchers write when they enter a topic, and whether they follow others into it as the herding account implies.

5.1 Newcomers propose methods

We registered the prediction that newcomer papers, whose authors face the most uncertainty, would more often be follow-on work: measurement papers that benchmark, collect data for or evaluate what others had proposed. Newcomer papers make up 73% of the 42,325 papers in 2019–2026 that have an author identifier. Against incumbent papers, the difference in the share of method papers is +3.71 pp [+2.61, +4.80] in discovery and +3.60 [+1.48, +5.71] on held-out papers; for measurement papers it is −1.70 [−2.46, −0.93] and −2.10 [−3.35, −0.85] (Table 5.1, Figure 5.1). Both contrasts are REPLICATED. Our registered prediction thus failed in the opposite direction, and the reversed hypotheses, registered before the held-out data were released to us, were supported on the held-out papers. All ten counted variants agree in sign, and the t-statistics (6.7 and −4.4) are too large to be explained by the inflated false-positive rate reported in Chapter 3. Team size accounts for much of the measurement gap: with log team size as a control, the gap shrinks to −0.54 [−1.11, +0.04] in discovery and −0.98 ($p = 0.10$) on held-out papers, whereas the method gap barely changes (+3.92 and +3.64).

Contrast	Discovery [95% CI]	Held-out [95% CI]	Label
Method paper	+3.71 [+2.61, +4.80]	+3.60 [+1.48, +5.71]	REPLICATED
Measurement paper	−1.70 [−2.46, −0.93]	−2.10 [−3.35, −0.85]	REPLICATED
Measurement paper, with log team size[a]	−0.54 [−1.11, +0.04]	−0.98 ($p = 0.10$)	REGISTERED VARIANT
Method: returning topic newcomer − incumbent[b]	+3.43 [+2.31, +4.55]	–	EXPLORATORY
Method: first-time team − returning topic newcomer[b]	+0.92 [−0.44, +2.29]	–	EXPLORATORY
Distance from earlier ICLR work, own work excluded (percentile points)	+4.33 [+3.44, +5.22]	–	EXPLORATORY
Entry into hot topics: incumbent share, hot − other topics[c]	−0.73 [−2.68, +1.22]	–	INCONCLUSIVE

TABLE 5.1: Entry contrasts: newcomer minus incumbent papers unless the row names other profiles, in pp within topic and year. Held-out tests are one-sided. [a]Registered supplementary variant, outside the support rule. [b]First-time team: no author has any earlier ICLR submission; returning topic newcomer: an author has one, in another topic. [c]No held-out test was registered.

5.2 Topic newcomers and first-time ICLR teams

Newcomer papers split into those from first-time teams (26.3% of papers; median team size 3) and those from returning topic newcomers (46.7%; median 5); incumbent papers make up the remaining 27.0% (median 5). Returning topic newcomers write method papers more often than incumbents (+3.43 pp [+2.31, +4.55]) at the same median team size, and the difference holds with team size controlled (+3.62). For first-time teams the difference is +4.35 [+2.82, +5.89], and the interval for the gap between the two newcomer profiles includes zero (+0.92 [−0.44, +2.29]). All three contrasts are EXPLORATORY. The method surplus, then, is associated mainly with entering a topic rather than with being new to ICLR or with working in a small team, and it appears in every period, including before 2023.

The measurement gap behaves differently. For first-time teams it is accounted for by team size (−2.24 [−3.24, −1.25] against incumbents, and +0.09 [−0.63, +0.80] with team size controlled); for returning topic newcomers part of it remains (−1.47, and −0.81 [−1.46, −0.16] with team size controlled). The gap appears only in 2024–2026 (first-time teams against incumbents: −2.75 [−4.07, −1.44]). In the FM era, then, measurement papers are concentrated among topic incumbents and larger teams. Team size is the only resource we observe; others, such as compute, may also matter. NLP

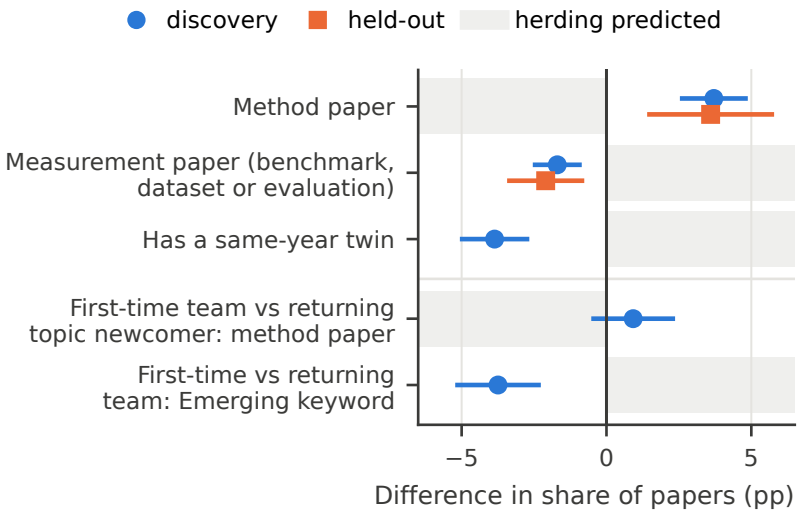


FIGURE 5.1: Entry contrasts within topic and year (pp), with 95% intervals: discovery (blue circles) and held-out (orange squares). Rows 1–2 compare newcomer with incumbent papers; they are registered and REPLICATED. Rows 3–5 are EXPLORATORY (discovery only). Row 3 compares papers by returning topic newcomers with incumbent papers, with first-time teams entered separately and with contribution-type, FM and team-size controls. Row 4 compares first-time teams with returning topic newcomers, and row 5 compares first-time teams with returning teams (all teams with earlier ICLR submissions) on the use of young, rising Emerging keywords. Grey shading marks the side that the herding prediction expected.

shows a parallel at the level of research trends, where tasks and methods lead and datasets follow [56].

5.3 Farther from earlier work

Newcomer papers lie farther from earlier ICLR work than incumbent papers. Within topic and year the difference is +5.90 percentile points [+4.95, +6.85]; it is +5.24 among method papers alone and +4.33 [+3.44, +5.22] when the authors' own earlier papers are excluded from the reference set. Papers by returning topic newcomers are less likely than incumbent papers to have a same-year *twin*, an independent submission at cosine similarity ≥ 0.70; with first-time teams controlled separately, the difference in twin share is -3.86 pp [-4.98, -2.74]. These results are EXPLORATORY. The follow-on prediction expects newcomer papers to lie closer to earlier work. Writing method papers could itself be imitation of the venue's dominant genre, since three quarters of ICLR papers are method papers, but newcomers' distance from earlier work argues against that reading. Neither contribution type nor embedding distance measures originality;

these results argue against newcomers following others but say nothing about whether their work is more original.

5.4 Later uptake of foundation models

Among 2023–2026 papers in topics where 5–50% of the previous year's papers built on FMs, the gap in FM-paper share between first-time teams and returning topic newcomers is −4.58 pp [−6.55, −2.60], widening to −14.71 [−19.82, −9.61] as adoption rises. For returning topic newcomers and incumbents the data show no clear difference. First-time teams also list *Emerging* keywords, which are young and rising, at a lower rate than teams with earlier ICLR submissions; the difference is −3.74 [−5.14, −2.35]. Both results are EXPLORATORY. Foundation models thus spread through teams with earlier ICLR work, as absorptive capacity predicts [30]. Network diffusion predicts the same order, however, so this contrast cannot tell the routes apart.

5.5 Open questions

Whether entrants crowd into hot topics remains open. The registered test of entry into hot topics, which compares the incumbent share in hot and other topics, is inconclusive (−0.73 [−2.68, +1.22]), and no held-out test was registered for it. None of the entry contrasts we measured favours the follow-on prediction.

Chapter 6

The Gate: Crowd or Record?

This chapter asks whether acceptance at ICLR favours papers that follow the crowd or papers anchored to their authors' own earlier work.

6.1 Is following the crowd associated with acceptance?

Table 6.1 and Figure 6.1 collect the tests at the gate. Of those in the figure, crowding was registered, title style had been explored before registration and the remaining tests were found post hoc in discovery; each held-out test was declared before any held-out estimate for its question was computed.

The registered crowding test predicted a penalty for crowded topics, and it is uninformative. Per log unit of crowding (the natural log of one plus the number of the topic's earlier papers), acceptance changes by −1.68 pp [−3.68, +0.32] in discovery and by −1.51 [−5.20, +2.19] on held-out papers, against an MDE80 of 4.8. In short, the data show no clear penalty or premium, although held-out premiums above +2.19 pp per log unit lie outside the interval.

The clearest sign of imitation in the corpus is in title style. The share of titles that open with a short coined name and a colon rose from 10.9% in 2017 to 44.6% in 2026. Within topic and year, with contribution-type, FM and team-size controls, such branded titles are accepted 1.74 pp [+0.37, +3.11] more often in discovery, but this estimate is not a test, since title style had been explored before registration. The declared held-out test

gives +1.76 [−0.82, +4.34] (one-sided $p = 0.089$); the held-out test is uninformative (below its MDE80 of 3.2), so the premium remains EXPLORATORY. Raw within-year gaps are larger (+3.3 in discovery and +4.0 on held-out papers); about half of each gap is explained by topic, contribution type, FM use and team size. Titles can also be revised after the decision, which could create such a premium.

FM papers are accepted 4.62 pp more often [+2.79, +6.44], with no sign of a difference between late and early adopters: the premium changes by +0.19 [−0.96, +1.34] per logit unit of the topic's prior FM adoption (that is, per unit of its log-odds). Both results are EXPLORATORY, and they fit efficient choice as well as imitation, if review rewards it. A topic's *edge* is its acceptance rate minus that of all other papers of the same year, using topics of the 2017–2020 map. The edge persists but shrinks towards the mean: among discovery papers, the 2018–2022 edge predicts the 2024–2026 edge with slope 0.38 [0.18, 0.58]. The held-out test asks a related but different question, namely whether the discovery edges for 2018–2022 predict the 2024–2026 edges of held-out papers; it gives a slope of 0.60 [0.32, 0.88] and is HELD-OUT SUPPORTED. A favoured topic thus keeps only part of its edge, as expected when decisions carry a large arbitrary component [59, 60].

Keyword timing shows at most weak associations, and we treat it as EXPLORATORY: keyword popularity had been explored against acceptance before registration, and both held-out intervals below include zero. On held-out papers, *Emerging* keywords (young and rising) are associated with higher acceptance than *Mainstream* keywords (old and flat), by +3.25 [−0.18, +6.69], and acceptance may fall after a keyword's share peaks (−5.2 [−11.1, +0.7]; −1.7 without the six keywords examined before this test was declared). Neither pattern shows the constant premium for popular keywords that imitation would predict if review rewarded it.

6.2 What is acceptance associated with?

The main pattern is the *far jump*: the lower acceptance of papers far from the team's earlier work. Among 20,430 decided 2023–2026 papers with an experienced author, acceptance at the 10th percentile of closeness to the team's own record is 5.44 pp lower than at the median in discovery (−5.44 [−7.10, −3.78]) and 6.09 pp lower on held-out papers (−6.09 [−9.32, −2.86]), with power = 0.89. This was the primary test declared

Contrast in acceptance	Discovery [95% CI]	Held-out [95% CI]	Label
Referenced to the team's own record			
Far jump: 10th percentile of $F_{\max}$ − median	−5.44 [−7.10, −3.78]	−6.09 [−9.32, −2.86]	HELD-OUT SUPPORTED
Same author, 0.1 closer in cosine similarity[c]	+0.55 [+0.21, +0.90]	+1.06 [+0.23, +1.89]	HELD-OUT SUPPORTED
One topic insider vs none[c]	+3.42 [+1.18, +5.66]	+3.88 [+0.28, +7.47]	HELD-OUT SUPPORTED
Referenced to others or the field			
Crowding, per log unit[a]	−1.68 [−3.68, +0.32]	−1.51 [−5.20, +2.19]	UNINFORMATIVE
Branded title[b]	+1.74 [+0.37, +3.11]	+1.76 [−0.82, +4.34]	EXPLORATORY
Change in FM premium per logit unit of prior adoption	+0.19 [−0.96, +1.34]	–	EXPLORATORY
Topic edge 2018–22 → 2024–26, slope[c]	0.38 [0.18, 0.58]	0.60 [0.32, 0.88]	HELD-OUT SUPPORTED
Emerging − Mainstream keyword[b,c]	+2.67 [+0.76, +4.58]	+3.25 [−0.18, +6.69]	EXPLORATORY
Distance from earlier ICLR work: least-distant fifth − middle fifth	−3.27 [−4.93, −1.60]	−0.13 [−3.50, +3.25]	UNINFORMATIVE

TABLE 6.1: Acceptance contrasts at the gate, in pp within topic and year (topic edges and keywords: against the same year's other papers). Rows differ in population and unit, so their sizes are not comparable. Held-out tests are one-sided. [a]Registered; predicted a penalty; MDE80 4.8. [b]Explored against acceptance before registration. For titles the discovery estimate is not a test (held-out MDE80 3.2); for keywords it is a declared post hoc test on a dimension already seen. [c]Holm-adjusted p across the six secondary held-out tests: topic-edge slope 0.019, edge prediction 0.026, same-author closeness 0.026, insider 0.052, Emerging keywords 0.063, same-author distance slope 0.22.

for the held-out papers, and it is HELD-OUT SUPPORTED. Adjusted acceptance rises with closeness up to the fourth quintile. The far jump remains with a newcomer control (−5.0), although closeness is correlated with newcomer status ($r = -0.57$).

Holding the author fixed, a paper 0.1 closer in cosine similarity to that author's earlier work is associated with acceptance 0.55 pp higher [+0.21, +0.90] in discovery and 1.06 pp higher [+0.23, +1.89] on held-out papers. This result is HELD-OUT SUPPORTED and survives a family-wise correction across the secondary tests. Controls for newcomer status and for co-authors' experience change it little (+0.48 and +0.50), though unmeasured differences in co-author quality could also bias it; a simulation run before the held-out test showed that such a confounder can shift the estimate. In a related within-author comparison, on a paper in a topic new to them, the same author's acceptance is 1.37 pp lower (−1.37 [−2.74, −0.004]; author fixed effects, EXPLORATORY).[1]

[1]Newcomer papers as a whole have 10.2 pp lower acceptance (−10.2 [−11.8, −8.7]) within topic and year (EXPLORATORY), but that gap mostly reflects who writes them: it shrinks once team size and the authors' ICLR experience are controlled, and shrinks further within the same authors.

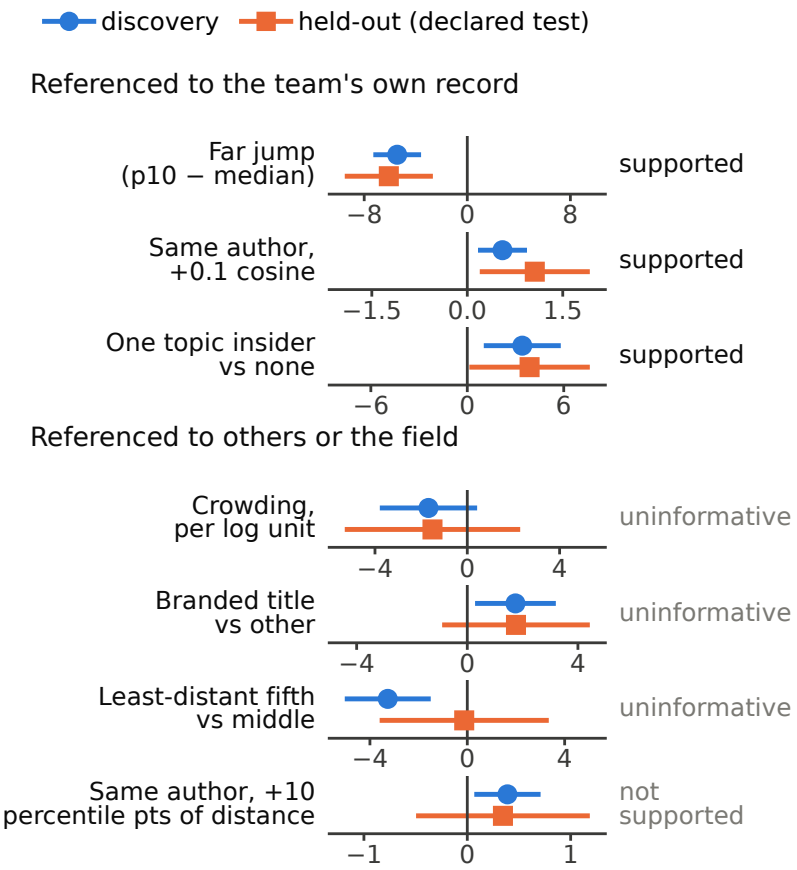


FIGURE 6.1: Gate tests by reference point: discovery estimates (blue circles) and declared held-out tests (orange squares), with 95% intervals. The held-out verdict is shown at right: 'supported' means HELD-OUT SUPPORTED; 'uninformative' means not supported and below the test's MDE80; 'not supported' marks a declared test without a registered MDE80, which we call inconclusive. Each row has its own scale: pp for the far jump (closeness at its 10th percentile, p10, minus the median), the insider, the branded title and the least-distant fifth; pp per log unit for crowding; and pp per 0.1 cosine similarity and per 10 percentile points of distance for the two same-author slopes. Because the scales differ, compare verdicts rather than sizes.

Having one topic insider on the team, rather than none, is associated with acceptance 3.42 pp higher [+1.18, +5.66] in discovery and 3.88 pp higher [+0.28, +7.47] on held-out papers, at fixed team size and number of experienced authors (HELD-OUT SUPPORTED). The result does not survive the family-wise correction (Holm-adjusted $p = 0.052$), and status explains it equally well: each additional insider adds only 0.58 pp more than each additional experienced author from another topic ($p = 0.16$; EXPLORATORY).

Distance from earlier ICLR work as a whole gives uninformative results. In the same SPECTER2 embedding space, papers in the least-distant fifth differ from those in the middle fifth by −3.27 [−4.93, −1.60] in discovery but by −0.13 [−3.50, +3.25] on held-out papers, a test with power = 0.47. The within-author slope for distance from earlier

ICLR work was not supported either, and with no MDE80 available it is inconclusive. Of the distance-based tests, only the associations with the team's own earlier work held up on held-out papers.

6.3 Interpretation

These associations all involve closeness to the team's own earlier work, as continuity predicts, but expertise, evaluator conservatism and visible status predict them as well [35, 36]. Double-blind review does not rule status out: where reviewers can see the authors, famous authors fare better [61], and preprints can reveal who wrote a paper [62].

Bibliography

[1] David Jurgens. The future of NLP may not be at NLP conferences: Scholarly migration patterns in natural language processing. arXiv preprint arXiv:2607.02416, 2026.

[2] Diana Crane. Fashion in science: Does it exist? *Social Problems*, 16(4):433–441, 1969. doi: 10.1525/sp.1969.16.4.03a00040.

[3] Zachary C. Lipton and Jacob Steinhardt. Troubling trends in machine learning scholarship: Some ML papers suffer from flaws that could mislead the public and stymie future research. *Queue*, 17(1):45–77, 2019. doi: 10.1145/3317287.3328534.

[4] Sara Hooker. The hardware lottery. *Communications of the ACM*, 64(12):58–65, 2021.

[5] Mostafa Dehghani, Yi Tay, Alexey A. Gritsenko, Zhe Zhao, Neil Houlsby, Fernando Diaz, Donald Metzler, and Oriol Vinyals. The benchmark lottery. arXiv preprint arXiv:2107.07002, 2021.

[6] Rishi Bommasani et al. On the opportunities and risks of foundation models. arXiv preprint arXiv:2108.07258, 2021.

[7] Sushil Bikhchandani, David Hirshleifer, and Ivo Welch. A theory of fads, fashion, custom, and cultural change as informational cascades. *Journal of Political Economy*, 100(5):992–1026, 1992.

[8] Abhijit V. Banerjee. A simple model of herd behavior. *Quarterly Journal of Economics*, 107(3):797–817, 1992.

[9] Eric Abrahamson. Managerial fads and fashions: The diffusion and rejection of innovations. *Academy of Management Review*, 16(3):586–612, 1991.

[10] Amos Tversky and Daniel Kahneman. Judgment under uncertainty: Heuristics and biases. *Science*, 185(4157):1124–1131, 1974. doi: 10.1126/science.185.4157.1124.

[11] Michelle Baddeley. Herding, social influences and behavioural bias in scientific research. *EMBO Reports*, 16(8):902–905, 2015. doi: 10.15252/embr.201540637.

[12] David Tran, Alex Valtchanov, Keshav Ganapathy, Raymond Feng, Eric Slud, Micah Goldblum, and Tom Goldstein. An open review of OpenReview: A critical analysis of the machine learning conference review process. arXiv preprint arXiv:2010.05137, 2020.

[13] Hazem Ibrahim, Talal Rahwan, and Yasir Zaki. Bias at the borderline: Who gets the benefit of the doubt in peer review? arXiv preprint arXiv:2607.26280, 2026.

[14] Sangkeun Jung, Goun Pyeon, Inbum Heo, and Hyungjin Ahn. What drives paper acceptance? A process-centric analysis of modern peer review. arXiv preprint arXiv:2509.25701, 2025.

[15] Gang Wang, Qi Peng, Yanfeng Zhang, and Mingyang Zhang. What have we learned from OpenReview? *World Wide Web*, 26:683–708, 2023. doi: 10.1007/s11280-022-01109-z.

[16] Binyan Xu et al. Reviewer scores are not comparable across research areas in ML peer review. arXiv preprint arXiv:2607.27209, 2026.

[17] Fan Huang. Catalyst papers in artificial intelligence research: A landscape on ICLR from 2017 to 2025. arXiv preprint arXiv:2607.05401, 2026.

[18] Rajiv Movva, Sidhika Balachandar, Kenny Peng, Gabriel Agostini, Nikhil Garg, and Emma Pierson. Topics, authors, and institutions in large language model research: Trends from 17K arXiv papers. In *Proceedings of the 2024 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies (Volume 1: Long Papers)*, pages 1223–1243, Mexico City, Mexico, 2024. Association for Computational Linguistics. URL https://aclanthology.org/2024.naacl-long.67.

[19] Honglin Bao and Kai Li. Persistence paradox in dynamic science. arXiv preprint arXiv:2506.22729, 2025.

[20] Ryan Hill, Yian Yin, Carolyn Stein, Xizhao Wang, Dashun Wang, and Benjamin F. Jones. The pivot penalty in research. *Nature*, 642(8069):999–1006, 2025. doi: 10.1038/s41586-025-09048-1.

[21] Daniele Rotolo, Diana Hicks, and Ben R. Martin. What is an emerging technology? *Research Policy*, 44(10):1827–1843, 2015. doi: 10.1016/j.respol.2015.06.006.

[22] Xue Li, Ciro D. Esposito, Paul Groth, Jonathan Sitruk, Balazs Szatmari, and Nachoem Wijnberg. Evaluation of unsupervised static topic models' emergence detection ability. *PeerJ Computer Science*, 11:e2875, 2025. doi: 10.7717/peerj-cs.2875.

[23] Rasul Khanbayov and Hasan Kurban. Topical phase transitions in artificial intelligence research: Large-scale evidence and an early-warning signature for emerging topics. arXiv preprint arXiv:2606.12828, 2026.

[24] Paul J. DiMaggio and Walter W. Powell. The iron cage revisited: Institutional isomorphism and collective rationality in organizational fields. *American Sociological Review*, 48(2):147–160, 1983. doi: 10.2307/2095101.

[25] David S. Scharfstein and Jeremy C. Stein. Herd behavior and investment. *American Economic Review*, 80(3):465–479, 1990.

[26] Eric Abrahamson and Lori Rosenkopf. Institutional and competitive bandwagons: Using mathematical modeling as a tool to explore innovation diffusion. *Academy of Management Review*, 18(3):487–517, 1993. doi: 10.2307/258906.

[27] Bernard Koch, Emily Denton, Alex Hanna, and Jacob G. Foster. Reduced, reused and recycled: The life of a dataset in machine learning research. In *Proceedings of the Neural Information Processing Systems Track on Datasets and Benchmarks*, volume 1, 2021. URL https://datasets-benchmarks-proceedings.neurips.cc/paper/2021/hash/3b8a614226a953a8cd9526fca6fe9ba5-Abstract-round2.html.

[28] James G. March. Exploration and exploitation in organizational learning. *Organization Science*, 2(1):71–87, 1991. doi: 10.1287/orsc.2.1.71.

[29] Toby E. Stuart and Joel M. Podolny. Local search and the evolution of technological capabilities. *Strategic Management Journal*, 17(S1):21–38, 1996. doi: 10.1002/smj.4250171004.

[30] Wesley M. Cohen and Daniel A. Levinthal. Absorptive capacity: A new perspective on learning and innovation. *Administrative Science Quarterly*, 35(1):128–152, 1990. doi: 10.2307/2393553.

[31] An Zeng, Ying Fan, Zengru Di, Yougui Wang, and Shlomo Havlin. Fresh teams are associated with original and multidisciplinary research. *Nature Human Behaviour*, 5(10):1314–1324, 2021. doi: 10.1038/s41562-021-01084-x.

[32] Mikko Packalen and Jay Bhattacharya. Age and the trying out of new ideas. *Journal of Human Capital*, 13(2):341–373, 2019. doi: 10.1086/703160.

[33] Michael Weisberg and Ryan Muldoon. Epistemic landscapes and the division of cognitive labor. *Philosophy of Science*, 76(2):225–252, 2009. doi: 10.1086/644786.

[34] Jason McKenzie Alexander, Johannes Himmelreich, and Christopher Thompson. Epistemic landscapes, optimal search, and the division of cognitive labor. *Philosophy of Science*, 82(3):424–453, 2015. doi: 10.1086/681766.

[35] Robert K. Merton. The Matthew effect in science. *Science*, 159(3810):56–63, 1968. doi: 10.1126/science.159.3810.56.

[36] Danielle Li. Expertise versus bias in evaluation: Evidence from the NIH. *American Economic Journal: Applied Economics*, 9(2):60–92, 2017. doi: 10.1257/app.20150421.

[37] James Coleman, Elihu Katz, and Herbert Menzel. The diffusion of an innovation among physicians. *Sociometry*, 20(4):253–270, 1957.

[38] Arman Cohan, Sergey Feldman, Iz Beltagy, Doug Downey, and Daniel Weld. SPECTER: Document-level representation learning using citation-informed transformers. In *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics*, pages 2270–2282, Online, 2020. Association for Computational Linguistics. doi: 10.18653/v1/2020.acl-main.207.

[39] Amanpreet Singh, Mike D'Arcy, Arman Cohan, Doug Downey, and Sergey Feldman. SciRepEval: A multi-format benchmark for scientific document representations. In

Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing, pages 5548–5566, Singapore, 2023. Association for Computational Linguistics. doi: 10.18653/v1/2023.emnlp-main.338.

[40] Caleb Ziems, William Held, Omar Shaikh, Jiaao Chen, Zhehao Zhang, and Diyi Yang. Can large language models transform computational social science? *Computational Linguistics*, 50(1):237–291, 2024. doi: 10.1162/coli_a_00502.

[41] Carolina Biliotti, Luca Verginer, and Massimo Riccaboni. Breaking new ground, reinforcing old gaps: Gender disparities in access to emerging research frontiers. arXiv preprint arXiv:2404.04707, 2024.

[42] Shachar Kaufman, Saharon Rosset, Claudia Perlich, and Ori Stitelman. Leakage in data mining: Formulation, detection, and avoidance. *ACM Transactions on Knowledge Discovery from Data*, 6(4):15:1–15:21, 2012. doi: 10.1145/2382577.2382579.

[43] Sture Holm. A simple sequentially rejective multiple test procedure. *Scandinavian Journal of Statistics*, 6(2):65–70, 1979.

[44] Brian A. Nosek, Charles R. Ebersole, Alexander C. DeHaven, and David T. Mellor. The preregistration revolution. *Proceedings of the National Academy of Sciences*, 115(11):2600–2606, 2018. doi: 10.1073/pnas.1708274114.

[45] Olmo R. van den Akker, Sara Weston, Lorne Campbell, et al. Preregistration of secondary data analysis: A template and tutorial. *Meta-Psychology*, 5, 2021. doi: 10.15626/MP.2020.2625.

[46] A. Colin Cameron, Jonah B. Gelbach, and Douglas L. Miller. Bootstrap-based improvements for inference with clustered errors. *The Review of Economics and Statistics*, 90(3):414–427, 2008. doi: 10.1162/rest.90.3.414.

[47] David Hall, Daniel Jurafsky, and Christopher D. Manning. Studying the history of ideas using topic models. In *Proceedings of the 2008 Conference on Empirical Methods in Natural Language Processing*, pages 363–371, Honolulu, Hawaii, 2008. Association for Computational Linguistics. URL https://aclanthology.org/D08-1038.

[48] Ashton Anderson, Dan Jurafsky, and Daniel A. McFarland. Towards a computational history of the ACL: 1980–2008. In *Proceedings of the ACL-2012 Special Workshop on Rediscovering 50 Years of Discoveries*, pages 13–21, Jeju Island, Korea, 2012. Association for Computational Linguistics. URL https://aclanthology.org/W12-3202/.

[49] Joel Klinger, Juan Mateos-Garcia, and Konstantinos Stathoulopoulos. A narrowing of AI research? arXiv preprint arXiv:2009.10385, 2020.

[50] David M. Blei and John D. Lafferty. Dynamic topic models. In *Proceedings of the 23rd International Conference on Machine Learning (ICML '06)*, volume 148 of *ACM International Conference Proceeding Series*, pages 113–120. ACM, 2006. doi: 10.1145/1143844.1143859.

[51] Maarten Grootendorst. BERTopic: Neural topic modeling with a class-based TF-IDF procedure. arXiv preprint arXiv:2203.05794, 2022.

[52] Sayash Kapoor and Arvind Narayanan. Leakage and the reproducibility crisis in machine-learning-based science. *Patterns*, 4(9):100804, 2023. doi: 10.1016/j.patter.2023.100804.

[53] Baruch Fischhoff. Hindsight ≠ foresight: The effect of outcome knowledge on judgment under uncertainty. *Journal of Experimental Psychology: Human Perception and Performance*, 1(3):288–299, 1975. doi: 10.1037/0096-1523.1.3.288.

[54] Sireesh Gururaja, Amanda Bertsch, Clara Na, David Gray Widder, and Emma Strubell. To build our future, we must know our past: Contextualizing paradigm shifts in natural language processing. In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*, pages 13310–13325, Singapore, 2023. Association for Computational Linguistics. doi: 10.18653/v1/2023.emnlp-main.822. URL https://aclanthology.org/2023.emnlp-main.822.

[55] Naomi Saphra, Eve Fleisig, Kyunghyun Cho, and Adam Lopez. First tragedy, then parse: History repeats itself in the new era of large language models. In *Proceedings of the 2024 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies (Volume 1: Long Papers)*, pages 2310–2326, Mexico City, Mexico, 2024. Association for Computational Linguistics. doi: 10.18653/v1/2024.naacl-long.128.

[56] Aniket Pramanick, Yufang Hou, Saif M. Mohammad, and Iryna Gurevych. A diachronic analysis of paradigm shifts in NLP research: When, how, and why? In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*, pages 2312–2326, Singapore, 2023. Association for Computational Linguistics. doi: 10.18653/v1/2023.emnlp-main.142. URL https://aclanthology.org/2023.emnlp-main.142.

[57] Richard S. Sutton. The bitter lesson. Blog post, March 2019. URL http://www.incompleteideas.net/IncIdeas/BitterLesson.html.

[58] Joan H. Fujimura. The molecular biological bandwagon in cancer research: Where social worlds meet. *Social Problems*, 35(3):261–283, 1988.

[59] Corinna Cortes and Neil D. Lawrence. Inconsistency in conference peer review: Revisiting the 2014 NeurIPS experiment. arXiv preprint arXiv:2109.09774, 2021.

[60] Alina Beygelzimer, Yann N. Dauphin, Percy Liang, and Jennifer Wortman Vaughan. Has the machine learning review process become more arbitrary as the field has grown? The NeurIPS 2021 consistency experiment. arXiv preprint arXiv:2306.03262, 2023.

[61] Andrew Tomkins, Min Zhang, and William D. Heavlin. Reviewer bias in single- versus double-blind peer review. *Proceedings of the National Academy of Sciences*, 114(48):12708–12713, 2017. doi: 10.1073/pnas.1707323114.

[62] Homanga Bharadhwaj, Dylan Turpin, Animesh Garg, and Ashton Anderson. De-anonymization of authors through arXiv submissions during double-blind review. arXiv preprint arXiv:2007.00177, 2020.